HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed **Python**-based ML models for classification and ranking tasks, achieving a **15% improvement** in precision on a 250k-record dataset.
- Built and maintained **ML training, inference, and deployment pipelines** using **Python** and containerization, ensuring reproducible model releases.
- Improved search relevance by implementing **embedding-based hybrid search** and **reranking strategies**, leading to a **22% increase** in click-through rates for ranked results.
- Designed and implemented evaluation pipelines, including the generation of **labeled datasets** and custom **metrics** to rigorously measure model quality and identify regressions.
- Refactored existing **Python** data processing scripts to optimize performance, reducing processing time for key feature pipelines by an average of **30%**.
- Debugged and resolved production issues impacting ML model performance, monitoring key indicators to ensure continued reliability and shipping improvements based on real-world feedback.
- Containerized ML inference services using **Docker** and deployed them to a **Kubernetes** cluster on **AWS**, managing configuration and resource allocation.

SKILLS

ML Concepts – Classification, Ranking, Retrieval, Personalization, Embeddings, Neural Architectures, Model Evaluation, Experimentation, Metrics

Programming Languages – Python, SQL

ML Frameworks & Libraries – PyTorch, JAX, scikit-learn, pandas, NumPy

ML Systems & Deployment – Production ML Systems, Inference Pipelines, Training Pipelines, Deployment Pipelines, Vector Search, Hybrid Search, Reranking

Cloud & Containerization – AWS, Docker, Kubernetes

PROJECTS

ML Model for User Preference Ranking

- Engineered a feature pipeline in **Python** and **SQL** to extract user interaction signals from a dataset of approximately 1.5 million records.
- Trained a **PyTorch**-based ranking model to personalize content recommendations, achieving a **0.82 AUC** on a held-out validation set.
- Developed a FastAPI service for model inference, achieving a p95 latency of **180ms** and integrating basic input validation.

Classification Model for Content Moderation

- Built a classification model using **scikit-learn** to identify and flag inappropriate content, trained on a dataset of 300k text entries.
- Developed an automated evaluation pipeline in **Python** that calculated precision, recall, and F1-score for each model iteration, enabling rapid iteration.
- Integrated the trained model into a staging environment, demonstrating its capability to process ~1,200 classification requests per hour with **92%** accuracy on test data.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)